

Storage Services

Prof. Arnab Mandal

Assistant Professor

CSE Department

Content

1.Object Storage	3
2. Block Storage.....	4
3. File Storage	5
4 . Amazon S3.....	6
5. Amazon S3 Storage class.....	7
6 . Storage Class Comparison.....	18
8. Amazon EFS.....	19
9. Amazon EBS	21
12.Comparison of Amazon Core storage Services	22
13. Lab : Creating and Amazon EBS volume.....	23
14 Lab : Hosting a static website using S3	25

Object Storage:

Object storage manages data as "objects" rather than files or blocks. Each object consists of the data itself, a variable amount of metadata (providing rich context), and a globally unique identifier.

How it works: Data is stored in a flat address space (a "bucket"). You do not navigate a hierarchy; you access objects via a unique ID or HTTP/REST API calls.

Best for: Unstructured data, massive scalability, cloud-native applications, and long-term backups/archives (e.g., photos, videos, logs, data lakes).

Key Advantage: Infinite scalability and rich metadata tagging.

Block Storage:

Block storage breaks data into fixed-size "blocks," each with a unique address. These blocks are stored independently of one another.

How it works: The storage system acts like a raw hard drive attached to a server. The operating system manages the file system on top of these blocks. Because the connection is low-latency, the OS can read/write data very quickly.

Best for: Databases, enterprise applications (ERP, CRM), and virtual machine file systems that require high performance and low latency.

Key Advantage: High performance and granular control; ideal for transactional workloads.

File Storage:

File storage organizes data into a hierarchical structure of files and folders, similar to how paper documents are stored in a physical filing cabinet. It uses a naming convention and a specific path to locate and access data.

How it works: Data is stored as a single piece of information within a folder. These folders are nested under other folders to create a directory. The system uses protocols like NFS (Network File System) or SMB (Server Message Block) to allow multiple users to access and share the same files over a network.

Best for: Local archiving, shared file systems, home directories, and content management where human readability and simple organization are prioritized.

Key Advantage: Highly intuitive and easy to use; provides a familiar interface for users to collaborate and manage data without specialized software.

Amazon S3

Amazon S3 is a highly scalable, object-based cloud storage service. Unlike block or file systems, it stores data as "objects" within containers called "buckets," providing a flat address space that is accessible via web APIs.

How it works: You upload files (objects) into a bucket. Each object is identified by a unique key and can hold massive amounts of metadata. Because it is built on a distributed cloud architecture, it provides massive scale and can be accessed from anywhere via HTTPS, without needing to mount it as a drive.

Best for: Storing unstructured data, hosting static websites, building data lakes for analytics, serving as a primary backup and disaster recovery target, and distributing media content globally.

Key Advantage: Virtually unlimited scalability with industry-leading durability ("11 nines"), allowing you to store petabytes of data cost-effectively while using lifecycle policies to manage costs automatically.

Amazon S3 Storage Classes

Amazon S3 is an object-based storage service that scales to support individual files up to **5 TB**. Data is stored in containers called "buckets" and managed via unique object IDs.

S3 Storage Classes

AWS provides various tiers to optimize costs based on your access patterns:

S3 Standard: The default for **frequently accessed** data; offers high performance and low latency.

S3 Intelligent-Tiering: Automatically optimizes costs by moving data between **frequent and infrequent** access tiers based on real-time usage.

Infrequent Access (IA) Classes: Designed for data that is rarely accessed but requires **rapid retrieval** when needed.

S3 Glacier Classes: Specialized for **long-term archiving**, offering the lowest storage costs with retrieval times ranging from milliseconds to 48 hours.

S3 Standard is the default, high-performance storage class designed for frequently accessed data. It delivers low latency, high throughput, and high durability across at least three Availability Zones (AZs).

Key Characteristics

Performance: Optimized for millisecond access and high-frequency requests.

Durability & Availability: Designed for 99.999999999% (11 nines) of durability and high availability.

No Retrieval Fees: You pay only for storage and requests, making it cost-predictable for active workloads.

Primary Use Cases

Cloud-Native Applications: Serving as the backend for dynamic, high-traffic applications.

Content Delivery: Storing images, videos, and media files for rapid distribution.

Big Data & Analytics: Providing high-speed access for data mining and real-time processing tasks.

Mobile & Gaming: Supporting high-performance assets that users interact with daily.

2.S3 Intelligent-Tiering is the industry-leading storage class for data with unknown, changing, or unpredictable access patterns. It automatically optimizes your storage costs by moving objects between tiers based on your real-world usage.

Key Characteristics

Automatic Optimization: It monitors access patterns and shifts objects between a "Frequent Access" tier and an "Infrequent Access" tier without manual intervention.

No Retrieval Fees: Unlike other cost-saving tiers, Intelligent-Tiering carries **no retrieval charges**, making it ideal for workloads where access frequency is volatile.

Operational Efficiency: It eliminates the need for manual analysis or complex lifecycle policies to manage costs as data ages.

High Reliability: It maintains the same **99.999999999% (11 nines)** of durability as S3 Standard, with an availability service level of **99.9%**.

3.S3 Standard-Infrequent Access (Standard-IA) is designed for data that is not accessed frequently but requires rapid, millisecond-level access when needed. It provides the same performance as S3 Standard but at a lower storage price, making it an excellent choice for long-term storage requirements.

Characteristics

High Performance: Delivers the same high throughput and low latency as S3 Standard.

Resiliency: Data is stored redundantly across multiple Availability Zones (AZs) to ensure high durability.

Reliability: Offers **99.999999999% (11 nines)** of durability and **99.9%** availability.

Cost Structure: Lower storage cost than S3 Standard, though it involves a per-GB retrieval fee and a 30-day minimum storage charge.

How to Identify Suitable Data

Before moving data to Standard-IA, consider these four key factors:

Access Frequency: The data should be accessed rarely—ideally once a month or less.

Access Latency: If your application requires instant access when a request is made, this is the right class.

Data Size: Objects must be at least 128 KB (smaller objects are billed as 128 KB).

Durability Needs: Essential for critical data like backups and disaster recovery logs that must be protected against hardware failure.

Primary Use Cases

Long-Term Backups: Storing snapshots and archive data that you only need to retrieve during an audit or emergency.

Disaster Recovery (DR): Keeping replicas of mission-critical files off-site so they can be recovered immediately if the primary system fails.

Old Content: Preserving legacy files or older versions of data that are rarely touched but must be kept for compliance.

4. S3 Glacier Instant Retrieval: High-Performance Archiving with Rapid Access

S3 Glacier Instant Retrieval is an archival storage class designed to offer the lowest-cost storage for data archiving, while providing high performance and flexibility. It delivers the fastest access to archive storage, with data retrieval times measured in milliseconds comparable to S3 Standard.

Key Characteristics of S3 Glacier Instant Retrieval:

- Data recovery occurs within milliseconds.
- Minimum object size is 128 KB.
- Availability is 99.9%.
- Durability is 99.999999999% (11 nines).

5. S3 One Zone-Infrequent Access: Cost-Optimized Storage for a Single Availability Zone

Unlike other S3 storage classes that store data redundantly across a minimum of three Availability Zones, S3 One Zone-Infrequent Access (One Zone-IA) stores data in a single Availability Zone. This design reduces costs by approximately 20% compared to S3 Standard-IA. It is an excellent option for storing secondary backup copies of on-premises data or data that can be easily recreated. Despite the lower cost, S3 One Zone-IA provides high durability, high throughput, and low latency comparable to S3 Standard.

Key Characteristics of S3 One Zone-Infrequent Access:

- Supports SSL (Secure Sockets Layer) for secure data transfer and encryption.
- Data is vulnerable to loss in the event of Availability Zone destruction.
- Availability is 99.5%.
- Durability is 99.999999999% (11 nines).

6. S3 Glacier Flexible Retrieval: Balancing Cost and Retrieval Flexibility for Archiving

S3 Glacier Flexible Retrieval offers lower-cost storage compared to S3 Glacier Instant Retrieval. It is an ideal solution for backing up data that needs to be accessed infrequently, typically a few times per year. Data retrieval takes minutes, balancing cost savings with reasonable access times.

Key Characteristics of S3 Glacier Flexible Retrieval:

- Supports free recoveries in high quantities.
- Data access may be impacted if an Availability Zone (AZ) is destroyed.
- Well suited for retrieving large data sets, making it optimal for backup and disaster recovery use cases.
- Availability is 99.99%.
- Durability is 99.999999999% (11 nines).

7. Amazon S3 Glacier Deep Archive: Extremely Low-Cost Long-Term Storage

The Glacier Deep Archive storage class is designed for secure, long-lasting storage of large amounts of data at a cost competitive with off-premises tape archival services. It offers very affordable pricing while enabling data restoration within approximately 12 hours. This makes it an ideal choice for users who need durable, highly secure, and cost-effective storage for massive data volumes over extended periods.

Key Characteristics of S3 Glacier Deep Archive:

- Provides highly secure storage.
- Data recovery takes up to 12 hours.
- Availability is 99.99%.
- Durability is 99.999999999% (11 nines).

Storage Class Comparison:

Storage Class	Durability (Design)	Availability (SLA)	Stored in...	Retrieval Time	Min. Storage	Retrieval Fee?
S3 Standard	11 nines	99.99%	3+ AZs	Milliseconds	None	No
S3 Intelligent-Tiering	11 nines	99.9%	3+ AZs	Milliseconds	None	No
S3 Standard-IA	11 nines	99.9%	3+ AZs	Milliseconds	30 days	Yes
S3 One Zone-IA	11 nines	99.5%	1 AZ	Milliseconds	30 days	Yes
Glacier Instant	11 nines	99.9%	3+ AZs	Milliseconds	90 days	Yes
Glacier Flexible	11 nines	99.99%	3+ AZs	Minutes to Hours	90 days	Yes
Glacier Deep Archive	11 nines	99.99%	3+ AZs	< 12 Hours	180 days	Yes

Amazon EFS: Amazon Elastic File System (EFS) is a serverless, "set-and-forget" file storage system designed to be shared by multiple Linux-based resources at once. If you are explaining this to your students, it is best to frame it as **Network Attached Storage (NAS)** for the cloud.

Key Characteristics of EFS

- **Highly Scalable:** Unlike EBS (where you must pick a size like 100GB), EFS is elastic. It grows and shrinks automatically as you add or remove files. You only pay for what you use.
- **Shared Access:** This is the "Killer Feature." While a standard EBS volume can usually only attach to one EC2 instance, a single EFS file system can be mounted by **thousands** of EC2 instances, on-premises servers, or containers (ECS/EKS) simultaneously.
- **Regional Availability:** EFS stores data across multiple Availability Zones (AZs) in a region. If one AZ goes down, your data remains accessible from another.
- **Linux Only:** It uses the NFSv4 protocol, meaning it is natively compatible with Linux but not typically used for Windows (which uses Amazon FSx for Windows).

EFS Storage Classes (Cost Optimization)

Just like S3, EFS has different tiers to save money:

- **EFS Standard:** For frequently accessed data.
- **EFS Infrequent Access (IA):** Much cheaper storage for files not accessed daily.
- **Lifecycle Management:** You can set a policy (e.g., "move files to IA if not opened in 30 days") to automate savings.

Amazon EBS

In EBS, data is stored in **blocks** of equal size. This makes it incredibly fast for databases and operating systems because the system can change one small block of data without rewriting the entire file (unlike S3).

Key Characteristics of EBS

- **Availability Zone (AZ) Locked:** An EBS volume is created in a specific AZ (e.g., ap-south-1a). It can only be attached to an EC2 instance in that **same** AZ.
- **Persistence:** Unlike "Instance Store" (temporary storage), data on an EBS volume survives even if you stop or terminate the EC2 instance (if configured to do so).
- **Snapshots:** You can take "point-in-time" backups of your EBS volumes. These are stored in S3 and can be used to recreate a volume in a different AZ or Region.
- **Elasticity:** You can increase the volume size or change the volume type (e.g., from HDD to SSD) on the fly without detaching it.

Comparison of AWS Core Storage Services

Feature	Amazon S3 (Object)	Amazon EFS (File)	Amazon EBS (Block)
Storage Type	Object Storage	Shared File System (NFS)	Block Storage
Common Analogy	A "Cloud Drive" (Google Drive/Dropbox)	A "Shared Office Folder" (NAS)	A "Personal Hard Drive" (SSD/HDD)
Accessibility	Over the Internet (via URL/API)	Attached to multiple EC2s (Linux)	Attached to one EC2 instance
Scalability	Unlimited & Automatic	Elastic (Auto-grows/shrinks)	Manual (Must resize volume)
Performance	High Latency (not for OS/DB)	Low Latency (shared)	Lowest Latency (Fastest)
Redundancy	Built-in (stored across 3+ AZs)	Built-in (stored across 3+ AZs)	Within a single AZ only
Ideal Use Case	Static websites, backups, media files	Shared code, user home directories	Databases, OS Boot volumes

Lab: Creating an Amazon EBS Volume

Use this walkthrough to show students how to add a "virtual hard drive" to a Linux or Windows server.

Step 1: Create the Volume

1. Log in to the **AWS Management Console**.
2. Navigate to **EC2 -> Elastic Block Store -> Volumes**.
3. Click **Create volume**.
4. **Volume Settings:**
 - **Volume type:** Select **General Purpose SSD (gp3)** (best price/performance).
 - **Size:** Enter a small size for the lab, e.g., 2 GiB.
 - **Availability Zone: CRITICAL!** It must be the same AZ as your EC2 instance (e.g., ap-south-1a).
5. Click **Create volume**.

Step 2: Attach to Instance

1. Select the new volume from the list.
2. Click **Actions** -> **Attach volume**.
3. Select your running **Instance ID**.
4. **Device name:** Keep the default (e.g., /dev/sdf).
5. Click **Attach volume**.

Lab: Hosting a Static Website using S3

This exercise demonstrates that S3 is more than just "storage"—it can act as a simple web server.

Step 1: Create the Bucket

1. Go to the **S3 Console** and click **Create bucket**.
2. **Bucket name:** Must be unique globally (e.g., my-cloud-class-2026).
3. **Object Ownership:** Ensure "ACLs disabled" is selected.
4. **Block Public Access: Uncheck** "Block all public access." (You will get a warning; check the box to acknowledge it).
5. Click **Create bucket**.

Step 2: Enable Static Website Hosting

1. Open your new bucket and click the **Properties** tab.
2. Scroll to the bottom to find **Static website hosting** and click **Edit**.
3. Select **Enable**.
4. **Index document:** Type index.html.
5. Click **Save changes**.
 - *Note the "Bucket website endpoint" URL that appears at the bottom.*

Step 3: Set Permissions (Bucket Policy)

1. By default, S3 is private. To make it a website, you must allow the public to "Read" your files.
2. Click the **Permissions** tab.
3. Under **Bucket policy**, click **Edit** and paste this JSON:

```
{  
  "Version": "2012-10-17",  
  "Statement": [  
    {  
      "Sid": "PublicReadGetObject",  
      "Effect": "Allow",  
      "Principal": "*",  
      "Action": "s3:GetObject",  
      "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"  
    }  
  ]  
}
```

(Replace YOUR-BUCKET-NAME with your actual bucket name.)

Step 4: Upload and Test

1. Create a simple file on your computer named index.html.
2. Upload it to the **Objects** tab of your bucket.
3. Open the **Endpoint URL** you found in Step 2. Your website is now live!

Key Takeaways:

- **Storage Types:** Understand the difference between Object Storage (S3—accessible via API/Internet), Block Storage (EBS—high-performance for OS/Databases), and File Storage (EFS—shared access for multiple servers).
- **Amazon S3:** Operates as a highly durable, unlimited Object Storage service that uses "Buckets" to store data globally; it is the foundation for data lakes and serverless web hosting.
- **S3 Storage Classes:** Optimize costs by choosing the right class based on access patterns, such as S3 Standard (frequent access), S3 Glacier Instant Retrieval (rare access but instant recovery), and S3 Deep Archive (lowest cost for long-term storage).
- **Amazon EBS (Elastic Block Store):** Acts as a virtual "Hard Drive" that is AZ-specific, meaning it must be in the same Availability Zone as the EC2 instance it is attached to.
- **Amazon EFS (Elastic File System):** Provides a managed, Regional file system (NFS) that can be mounted simultaneously by thousands of EC2 instances, making it ideal for shared application code or content.

- **EBS Lab Essentials:** When creating a volume, you must match the Availability Zone of your instance; after attaching, you must manually format and mount the drive within the OS to make it usable.
- **S3 Website Hosting:** Transform a bucket into a Static Website by enabling the hosting property, uploading an index.html file, and applying a Public Read bucket policy to allow internet access.

Parul[®]
University

NAAC
GRADE **A++**



<https://paruluniversity.ac.in/>

